

Characterising Toxicity in Language Models

TI3150 - Capstone applied AI Project Plan
Group 12

November 2024

1 Background and Project Aim

Background

Large Language Models (LLMs) are becoming increasingly sophisticated, producing outputs that are not only coherent but also impactful in shaping digital interactions. As these models grow in size and capability, their influence expands [1].

Generative AI, while impressive, operates by predicting the next word in a sequence based on patterns learned from vast datasets. This results in no true understanding of meaning or context behind its outputs. Consequently, this lack of comprehension and judgment becomes problematic when LLMs generate toxic content, as they often fail to recognize appropriate or inappropriate responses on their own. This issue becomes even more concerning in scenarios where developers, users, or stakeholders disregard the ethical implications of these outputs or fail to take responsibility for mitigating harm.

Language, being highly context-dependent, makes the identification and mitigation of toxic outputs a particularly challenging task. When characterizing these model failures, we encounter two main challenges: benchmarking toxicity and explaining why the model generated a toxic output.

Although defining toxic output through benchmarking is crucial for identifying failures, it is not sufficient to understand the underlying causes of toxicity. Benchmarks provide representative examples of toxic content, but do not inherently explain the mechanisms behind the generation of such outputs. To address this gap, we must go beyond the benchmarks and dive into explainability techniques.

Explainable AI (XAI) has long been a vital area of research, undergoing a renaissance with works like LIME and SHAP, which provide algorithmic explanations. Despite advances in generative AI, XAI remains essential for understanding and mitigating the toxic output produced by LLM. This is particularly important because LLM outputs can often be inconsistent, overly confident, or prone to hallucinations. Given the societal impact of language toxicity, ensuring transparency in model decision making is crucial.

Project Aim

The aim of this project is to systematically characterize toxicity in Language Models (LLMs), with a focus on understanding when and why toxic outputs are generated. By combining benchmarking techniques with explainability methods, this work seeks to provide a clearer picture of the conditions under which LLMs fail and the underlying mechanisms driving these failures.

Dataset

Decoding Trust is a project aimed at providing a thorough assessment of trustworthiness in Generative Pretrained Transformer (GPT) models [2]. The project provides datasets to test models on a variety of aspects of trustworthiness, including toxicity. The prompts concerning toxicity are a subset of Real-ToxicityPrompts [3]. The dataset consists of prompts, along with their toxicity score, calculated using PerspectiveAPI [4], a widely used toxicity-scoring tool developed by Google Jigsaw. This data will be used in this project to analyse models, as well as to analyse the features in prompts that solicit toxic responses.

Model Selection

Not all LLMs are created equal, there are many different architectures an LLM can be built with, which come with its own quirks. For this reason part of the project will look into a number of models to find out when various models generate toxic content, and investigate the level of toxicity generated by various models. This project will only look into models that can be run locally, are from different research groups, and are the newest and most advanced models readily available as of December 2024. The models that were selected for this project are: MistralAI's Mixtral 8x7B [5], Meta's llama3 [6], Google's Gemma [7], and BigScience's BLOOM [8].

2 Requirements

2.1 Comparing Toxicity Across LLMs

RQ1: *How prone are Mixtral 8x7B, llama3, Gemma, and BLOOM to generate toxic outputs when prompted?*

- **MUST:** Compare the outputs of Mixtral 8x7B, llama3, Gemma, and BLOOM by testing their toxicity given the same prompts. We use Perspective API to generate the toxicity scores.
- **SHOULD:** Review the toxic output to understand why they are toxic. Study what kind of words lead to a higher toxicity score.
- **COULD:** Create a guideline for humans to decide when an output can be considered toxic.

2.2 Lexical Analysis on Toxic Inputs

RQ2: *What are lexical features of prompts that lead Mixtral 8x7B, llama3, Gemma, and BLOOM to generate toxic outputs?*

- **MUST:** Identify toxic outputs by setting a toxicity score threshold and filtering the model outputs accordingly.
- **MUST:** Explain why, and visualise how, the outputs are toxic by using Captum that highlights which parts of the input caused the toxic result.
- **MUST:** Analyse the specific words or tokens in the input that contributed to toxicity. Focus on their meaning and lexical features.
- **COULD:** Design a clear process for humans to evaluate whether certain prompts are likely to lead to toxic outputs.

2.3 Syntactical Analysis on Toxic Inputs

RQ3: *Which syntactic structures of prompts lead the three selected LLMs to generate toxic outputs?*

- **COULD:** Study the sentence structure of the inputs that caused toxic outputs. Use tools like dependency parsing and constituent parsing to analyse how words relate to each other.
- **COULD:** Combine findings from this syntactical analysis with the lexical analysis of RQ2 to obtain a final evaluation process for humans, formulated in RQ2.

2.4 Wont's

- Design a formula/tool to detect toxicity.
- Create an application to predict the likelihood that toxic content is generated, given a prompt.

3 Validation

3.1 RQ1: Comparative Toxicity Analysis of LLMs

- **MUST: Quantitative Validation**

This must is completed when all the outputs, given the prompts of the *DecodingTrust* dataset (<https://decodingtrust.github.io/>), of Mixtral 8x7B, llama3, Gemma, and BLOOM are compared and the consistency of their outputs is scored. To measure toxicity, the authors of *DecodingTrust* use established benchmarks like toxicity classifiers and human evaluation[2]. They rely on metrics such as the percentage of toxic outputs, the severity of toxicity as measured by scales, and the context in which toxicity occurs. Additionally, they assess unintended biases in toxicity classifications, such as differential toxicity scores based on sensitive attributes.

Initial Toxicity Scoring:

We will use established toxicity classifiers and the Perspective API to assign toxicity scores to each output. The Perspective API provides categories such as TOXICITY, SEVERE_TOXICITY, IDENTITY_ATTACK, INSULT, PROFANITY, and THREAT, each scored between 0 (not toxic) and 1 (highly toxic)[4]. Based on these scores, we will identify the 50 most toxic outputs for each LLM. These are the outputs with the highest average toxicity or with notably high scores in any single toxicity category.

- **SHOULD: Review toxic output** This should be completed when five independent human evaluators have reviewed the 50 high-toxicity outputs from each model. In their reviews, each evaluator will:
 - Identify language triggers, such as repeated words, phrases, or rhetorical styles that appear in highly toxic outputs.
 - Examine contextual factors, including the presence of hate speech, identity attacks, threats, or other forms of toxicity.
 - Detect potential biases by observing any patterns that suggest systematically higher toxicity scores for specific demographic or identity groups.

After all evaluators have completed their assessments, their observations will be compiled into a single dataset, documenting language triggers, contextual elements, and any signs of bias or harmful content. By comparing the findings across all five evaluators, we gain a deeper understanding of the factors that most significantly contribute to elevated toxicity scores.

- **COULD: Qualitative Validation** The findings on why the outputs receive a high toxicity score are distilled into a guideline. This guideline helps humans to either *prevent* toxic outputs or *reduce* their toxicity by illustrating which aspects of the input cause elevated toxicity. In addition, these findings are compared to existing literature on toxicity prevention and reduction, thereby situating our recommendations within broader research on mitigating harmful content.

3.2 RQ2: Lexical Analysis of Toxic Inputs

- **MUST: Toxic Output Subset** We will use Perspective AI to generate a toxicity score for a range of LLM outputs. We then look at the toxicity scores Perspective API assigned to these outputs, and based on the distribution of the toxicity scores we pick a threshold for what scores are classified as toxic and non toxic. When we have found this threshold this must is completed.
- **MUST: Attribution-based Explanations** This must is completed if it is possible to visually show which inputs contribute most to generating toxic outputs. We plan to look into Integrated Gradients or Saliency maps, which maps what part of the input contributed most to the generated output, and Shapley Value Sampling, which maps what input features contribute the most to the output.
- **MUST: Lexical Analysis** This must is completed when we know what lexical features can lead to toxic outputs. Some lexical features we will look at are sentiment features (good, bad, terrible, amazing, etc.) and lexicon-based Features, so words from our input that occur in a given set of toxic words from huggingface.
- **COULD: Human Evaluation Protocol** This is completed if there are relevant findings that are added to the previously defined guideline. It may be possible that there are no relevant findings that can be added.

3.3 RQ3: Syntactic Analysis of Toxic Inputs

- **COULD: Syntactic Analysis** This could be completed when parsing methods give insight into how inputs that lead to toxic outputs are structured. Since parsing methods provide grammatical understanding, we will look at the grammatical structure of the text and the relation between words.
- **COULD: Integration with Lexical Analysis** This could be completed when the syntactic and lexical findings are combined and represented in a guideline (see RQ2).

4 Risk Analysis

4.1 RQ1: Comparative Toxicity Analysis of LLMs

- **MUST: Quantitative Validation** *RISK*: Low *IMPORTANCE*: High
We expect that we will be able to qualitatively compare the outputs of the four chosen LLMs and

calculate their toxicity scores. We already have some of the chosen LLMs running and providing outputs, and we also already can generate a toxicity score using Perspective API, so to complete the must we only have to combine the two.

- **SHOULD: *RISK*: Medium *IMPORTANCE*: High**
It is very important to understand why outputs are classified and what the words are that explicitly make the output toxic. Using Integrated Gradients or Saliency maps and Shapley Value Sampling with Captum will give us great insight in why the outputs are toxic. However due to hardware constraints and how hard these methods are to compute it is not very clear if we will be able to fully run the attribution methods.
- **COULD: Qualitative Validation *RISK*: High *IMPORTANCE*: Low**
Making a guideline for humans would be very interesting but it is not a priority and falls therefore outside the primary scope of this project. Since this is not very important for the project and it could be quite time-consuming, we do not think we will be able to complete this requirement.

4.2 RQ2: Lexical Analysis of Toxic Inputs

- **MUST: Toxic Output Subset *RISK*: Low *IMPORTANCE*: High**
It is important to characterise the outputs we consider toxic or not. For this, we have to set a threshold and filter the outputs accordingly to build a dataset of toxic outputs. We use perspective API for this, and already have the API running and giving a toxicity score to its input. We only have to give it a lot of generated inputs. We think we will complete this must.
- **MUST: Attribution Explanations *RISK*: Medium *IMPORTANCE*: medium**
Highlighting and visualizing what lexical parts make an output toxic is important for understanding how our model works and explaining it. Computing these attribution methods requires a lot of GPU power. As we only have limited hardware and limited time to use high-end GPU's we are not sure whether we will be able to compute this must.
- **MUST: Lexical Analysis *RISK*: Medium *IMPORTANCE*: High**
By analysing the lexical features of the output, we further understand the model. Because we do not have a lot of experience with language tools, we think we will have some difficulties completing this requirement.

4.3 RQ3: Syntactic Analysis of Toxic Inputs

- **COULD: Syntactic Analysis *RISK*: Medium *IMPORTANCE*: Low**
By analysing the syntactic features of the output, we further understand the model. Because we do not have a lot of experience with language tools, we think we will have some difficulties completing this requirement.
- **COULD: Integration with Lexical Analysis *RISK*: High *IMPORTANCE*: Low**
Tying together the lexical and syntactical analysis creates a better overview of the way the model works, and when and why an output is toxic. However, this is a low priority task, making it less likely that we will complete this requirement.

5 Teamwork

5.1 Decision taking

During the execution of the project, a number of decisions will have to be taken. To ensure transparency, a central record of important decisions will be maintained. This will be used to maintain an overview on what decisions were taken when, and note the most important justifications. An entry to this record looks like section 5.1.

Date	Decision	Reasoning
The date the decision was taken	a description of the decision	A short justification for the decision

5.2 File- and Version management

There are multiple types of files that will be handled during the project. Each of these requires its own way of file and version management.

Keeping all code in one central place is essential to ensure smooth collaboration. Furthermore, having a good version control system makes it possible to go to previous versions if something breaks and see who wrote what. For this reason, we will be using GitHub to save all our code.

To ensure that anyone could use the code and reproduce the findings of the project, proper documentation is needed. This documentation will be saved, along with the code, on GitHub.

The reports for this project will be written in $\text{\LaTeX}$ using Overleaf. This makes it possible for all group members to work together on the report at the same time and provides a version history feature.

Miscellaneous documents will be saved to a private drive. As these documents are necessary during the execution of the project, it is essential that all team members have access to these files. Miscellaneous files include: presentations, minutes of meetings, task assignment/current planning, and many more.

5.3 Code of Conduct

6 Planning

The planning includes a list of various deadlines and the pace the project should follow. This is a crude listing of general tasks, and does not yet include a very specific breakdown of these tasks. Furthermore, no assignment of tasks has been made yet, as the team feels it to be preferable to determine this on the go. This clearly has its risks, but also some more flexibility/responsibility, which the team agrees would add to the quality of the final product.

Week	Date	Task
Week 4	Wednesday, 4th December 2024	Deadline: Project Plan
		Decide on LLMs to use for analysis
Week 5-6		Explore Dataset
		Exams: DL and NLP
Week 7	Monday, 6th January 2025	Deadline: Revised Project Plan
	Friday, 10th January 2025	Ethics Reflection 1
		RQ1 Qualitative analysis(must) and human evaluation(should) completed
		Setup Lexical-(RQ2) and syntactical(RQ3) analysis
		Report findings on RQ1
Week 8		Lexical(RQ2)(must) and Syntactical(RQ3)(should) analysis
		XAI with Captum(must)(RQ2 and RQ3)
Week 9	Friday, 24th January 2025	Deadlines: Ethics Reflection 2, Final Code, Final Report
		Complete RQ2 and RQ3(must/should)
Week 10	Wednesday, 29th January 2025 (9:00)	Final Presentation

References

- [1] Y. Hao, H. Song, L. Dong, *et al.*, *Language models are general-purpose interfaces*, 2022. arXiv: 2206.06336 [cs.CL]. [Online]. Available: <https://arxiv.org/abs/2206.06336>.
- [2] B. Wang, W. Chen, H. Pei, *et al.*, *Decodingtrust: A comprehensive assessment of trustworthiness in gpt models*, 2024. arXiv: 2306.11698 [cs.CL]. [Online]. Available: <https://arxiv.org/abs/2306.11698>.
- [3] S. Gehman, S. Gururangan, M. Sap, Y. Choi, and N. A. Smith, *Realtoxicityprompts: Evaluating neural toxic degeneration in language models*, 2020. arXiv: 2009.11462 [cs.CL]. [Online]. Available: <https://arxiv.org/abs/2009.11462>.
- [4] A. Lees, V. Q. Tran, Y. Tay, *et al.*, *A new generation of perspective api: Efficient multilingual character-level transformers*, 2022. arXiv: 2202.11176 [cs.CL]. [Online]. Available: <https://arxiv.org/abs/2202.11176>.
- [5] A. Q. Jiang, A. Sablayrolles, A. Roux, *et al.*, *Mixtral of experts*, 2024. arXiv: 2401.04088 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2401.04088>.
- [6] A. Grattafiori, A. Dubey, A. Jauhri, *et al.*, *The llama 3 herd of models*, 2024. arXiv: 2407.21783 [cs.AI]. [Online]. Available: <https://arxiv.org/abs/2407.21783>.
- [7] Gemma Team: T. Mesnard, C. Hardin, *et al.*, *Gemma: Open models based on gemini research and technology*, 2024. arXiv: 2403.08295 [cs.CL]. [Online]. Available: <https://arxiv.org/abs/2403.08295>.
- [8] BigScience Workshop: T. L. Scao, A. Fan, *et al.*, *Bloom: A 176b-parameter open-access multilingual language model*, 2023. arXiv: 2211.05100 [cs.CL]. [Online]. Available: <https://arxiv.org/abs/2211.05100>.